

Inbodied Interaction 102: Exploring Neuro-Physio Pathways for Self-Tuning

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ABSTRACT

In Inbodied Interaction 101, we considered the Physiology and Anatomy of the body via three associated interactions that reflect an **inbodied state**: 1. inbodied *adaptation* in response to the in5 and C4 over Time and Context in order to maintain 2. *homeostasis* via 3. *metabolism*. We called this adaptation process “tuning.” In 102 we build on this foundation to consider the physiology tuning. In particular we will look at a series of inbodied interactions: the neuro-endocrine system interaction with the organ systems that cue adaptive responses from genetic signals to fat metabolism; the autonomic nervous system and the limbic system’s interactions that affect volitional/non-volitional interaction. We will introduce the components of the brainstem, basal nuclei and cerebellum that support interoception around self-Tuning. Within this framing, we will look at the strengths and limits of non-invasive measures of these processes (eg, HRV, EEG, blood oxygen saturation, qualitative responses). Outcomes will familiarity with how we function as inbodied complex systems, with worked examples of how the physiology of tuning can be translated into interactive designs to support health, wellbeing, performance in new ways.

CCS CONCEPTS

• **Human-Centered Computing - Human Computer Interaction (HCI);**

KEYWORDS

Inbodied, Wellbeing, Human Performance, neurology, endocrinology, hormones, in5, c4, circumbodied, autonomic, breathing, brainstem, cerebellum

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1 INTRODUCTION

This course is a deeper dive into the physiology of *Inbodied Interaction*[7, 8]. In Inbodied Interaction 101, we framed the human body

as site of constant adaptation. To engage with this complex system of complex systems, we considered the Tuning Model. This model includes the 5 essential, semi-volitional practices of Move, Eat, Engage, Cogitate, Sleep (the in5 MEECS), the 4 constant parameters of Air, Light, Gravity and Microbiome (the circumbodied C4), and the variables of Time and Context that are always affecting this dynamic adaptation. $Tuning = (in5+c4)/T+C$. We considered how these components can be used deliberately in design along continua of insourcing to outsourcing, raw to cyborg, discomfort to ease.

In Inbodied Interaction 102, we open the design space further to consider more specifically the *inbodied pathways of tuning*. In other words, we will look a little deeper into the physiology of *how* Tuning operates, via three increasingly specific lenses: inbodied systems, signals, and integration. Through these lenses we will consider the interaction of the in5 with the organ systems of the body generally, the nervous system more particularly from senses to signals, and then areas of the brain other than the cortex such as the basal nuclei, cerebellum, midbrain pons and medulla that integrate these processes for action.

From these perspectives, we will have the foundations to consider the strengths and limitations of increasingly common measures in HCI from Heart Rate Variability to EEG, to practices like mindfulness, to consider how they may be re-situated to help support the components we touched on in 101 for Tuning from insourcing to outsourcing, discomfort to ease, raw to cyborg.

2 BENEFITS

Physiology is a vast topic, but anyone who wishes to design to *help* a body perform better, will only do that job better by understanding fundamentals of how that system functions – what its requirements are; what can make success less effortful and more likely and effective. The goal of this two unit CHI course is therefore to give participants a functional orientation into the physiology of the body from an inbodied interaction perspective: students will learn how to *align their designs*, and the evaluation of those designs with how the body itself works.

As part of this orientation, students will also leave this course with a new suite of tools to explore inbodied interaction that is, knowledge of how the body’s internal processes work optimally to inform design of tools to support and enhance that performance, including cognitive, physical, emotional, and social interaction. Just as HCI has been exploring questions informed by psychology around physical-mental processes like attention, breathing, and mindfulness (e.g., [10]), students will gain clear examples of how to ask new questions around newer-to-hci processes – such as the role of the midbrain pre cognition, and for interoception.

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We draw upon validated assessments from sports science, like visual acuity and peripheral awareness around cognitive performance and stress. We also introduce functional neurology measures for cerebellum, mesencephalon and insular cortex, for body awareness, stress, strength and social development. We will look at how these assessments complement and extend more familiar measures like HRV particularly for HCI. Beyond learning and practicing these new measures, participants will also gain new, deeper knowledge of neuro-physical processes that, while critical to our wellbeing, are as yet still less explored in HCI.

Course participants, using these tools for their future work in HCI, have the opportunity to become HCI path breakers and leaders in modeling for the community how these novel-to-hci assessments open up new conceptualizations of designs to support all aspects of our inbodied and embodied lives in terms of understanding the neurological and physiological pathways they both utilize and reveal.

3 INTENDED AUDIENCE

This course is designed for anyone who designs systems to aid human potential, performance, wellbeing and who wants to explore and open the possibilities of their design space, to help people build the knowledge, skills and practice we need as individuals and within groups to thrive together.

3.1 Prerequisites

This course is open to participants from the **Inbodied Interaction 101 course**, from the Inbodied Interaction Summer School. If you are interested in this course without having these pre-requisites, please make a case to the organizers at [II-102@nopain2.org]

4 CONTENT

To make the material in this course as useful and practicable as possible, we will use the in5's MOVE as our orientation point for the physiology of adaptation. MOVE is the first of the semi-volitional in5 MEECS we discuss in Inbodied 101. As a 101 review, we will first consider the functional properties of MOVEMENT relative to homeostasis, metabolism and adaptation. We will then situate movement relative to the three lenses of organs, signals, and integration, further detailed below.

1. **Lens One: Organs.** We will consider each of the 11 organ systems relative to the Tuning demands of movement for adaptation via metabolism to support homeostasis. 11 organ systems in the body not only supports movement, but perhaps more critically in terms of contemporary interactive design, requires movement for healthful homeostasis. Recent research has shown that we are constantly needing to reevaluate how we define these interactions. For example, work on skeletal processes shows bone may be involved in energy metabolism and cognition [1]. Likewise work around the fascia is suggesting it may be a "new" organ system itself [3]. Each of these systems adapts, or reshapes its cellular tissue, based on interactions like movement, with it. From just this overview of organ system interaction alone, we will be able to ask profound questions about evaluation of current designs from work to infrastructure as fit for purpose.

2. **Lens Two: Signals.** Moving has many volitional or conscious components, that relate to adaptations we often measure, from body composition to skills acquisition. We see these as outcomes of conscious deliberate practice. And yet, most of our movement processes happen without conscious attention. There is a constant interplay between conscious and the autonomic processes that support conscious action – and that significantly influence conscious decisions and actions. These processes – whether we are aware of them or not – are all enabled and affected by the signals from the nervous and endocrine systems. The body's capacity internally to respond to these signals influences how well our organs respond to adaptive cues, including how well we build new patterns and skills. These signals have particular dimensions from their direction of travel in the body, their speed along a given pathway, to their types of cellular or genetic interactions. We'll consider research [4] that is blurring the distinction between volitional and autonomic processes, and how these affect our health performance.
3. **Lens Three: Integration.** Within the nervous system, the autonomic system is constantly monitoring and mediating the signals from the neuro-endocrine systems in order to maintain homeostasis. When we move, we are integrating patterns and making decisions that require us to interpret our context for what is salient, and then execute responses to that event [5, 6]. This section considers the processes that support that complex interaction of signal perception to evaluation to action, introducing sub-cortical interactions via the brainstem, cerebellum and basal ganglia in particular.

From this overview of inbodied physiology of adaptation, we will look at how we can use current and evolving measures to assess our designs to support self-tuning in particular. We will consider the following questions:

1. **What are we measuring?** As is typically the case with physiological measures more generally, we are looking at *change* (e.g., heart rate *increases* or *decreases*). This section presents fundamental models to explain the underlying causes of these state changes to help designers better understand or model intervention effects, highlighting key characteristics of these processes such as whether they are mainly metabolic or physiologic in nature, differentiating between chronic and acute states, identifying key factors affecting types of state changes, and discussing relations between organ systems like endocrine, nervous, and enteric.
2. **How are we measuring?** What makes this an interesting and challenging area of study is that directly measuring changes in these bodily states is often infeasible, and therefore, many commonly used measures are actually proxies for these associated states. For example, HRV can be used as a proxy for breathing depth and recovery. Similarly, saccades – a visual assessment – provide insight into thalamus activity which can give us insight into neural pathway stimulation, and inform design decisions that could help improve, for example, attention, strength, sleep, etc. In this section, we will present and discuss the various proxy measures that



Figure 1: “Pencil Pushup” for convergence can be assessed anywhere, anytime. We will cover this in the course.

can be used to gain insight into internal bodily state, focusing specifically on implications for design, and trade-offs between invasiveness, ease of measurement, accuracy, and robustness to external factors.

3. **HCI examples of autonomic measures:** We review the role of the autonomic nervous system and consider key measures:
 - a. HRV as a proxy for breathing, as a proxy for neurological state dominance, and as a proxy for stress embodied in peripheral design for breathing application [9].
 - b. Peripheral vision as a proxy for peripheral awareness, as a proxy for EEG state, and as a controller for eBike acceleration [2].
4. **New Brainstem / Cerebellar Assessment Sampler.** This section will introduce a new-to-hci area of the brain – the midbrain and cerebellum – for consideration in HCI. New light weight measures and evaluation include: eye tracking, rapid alternating movements, air hunger challenges, vibration.

5 INSTRUCTORS BACKGROUNDS

m.c. is a professor of human performance and computer science. She leads the WellthLab in Human-Systems interaction, where the mission is to use science, engineering and design to explore how interactive technology can #makeNormalBetter @scale for all.

Aaron is a PhD student in Computer Science/HCI at UNB. His work focuses on rigorous interrogation of the use of HRV as a measure of the efficacy of breathing rate in particular, and the connection of breathing depth in cognitive performance. He is interested in developing cognitive assessment test suites for personal self-assessment of health interventions.

Josh leads HCI research at IBM Research, Australia, and has published particularly in exertion games, and is a founder of the Body as a Starting Point workshop series – he leads work on the use of Peripheral Awareness for eBike acceleration control.

6 PRACTICAL WORK AND RESOURCES

Given the online nature of the course, lecture with Q&A, and demonstration will be the main interaction, with breaks to test the mid-brain/cerebellar assessments as breaks after each section. We will also discuss together how they can be applied in a design context. Eg: understanding about the connection between vision, fatigue and balance may help applications move away from trying to correct posture before assessing vision. This approach has considerable implication for innovative screen designs.

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